A comparison of different types of blockchains

| Functionality | Public | Private |
|---|---|---|
| Access | Open read/write | Permissioned read and/or write |
| Speed | Slower | Faster |
| Security | Proof of work, proof of stake, other consensus mechanisms | Pre-approved participants |
| Identity | Anonymous | Known identities |
| Asset | Native asset | Any asset |
| Run | Anyone can run a BTC/LTC full node | Anyone cannot run a full node |
| Transaction | Anyone can make transactions | Anyone cannot make transactions |
| Audit | Anyone can review/ audit the blockchain | Anyone cannot review/audit the blockchain |
| Participants | Permissionless (anonymous) | Permissioned (identified and trusted) |
| Consensus | Proof of work, proof of stake | Voting or multi-party consensus algorithm |
| Transaction approval frequency | Long (10 minutes or more) | Short (100x msec) |

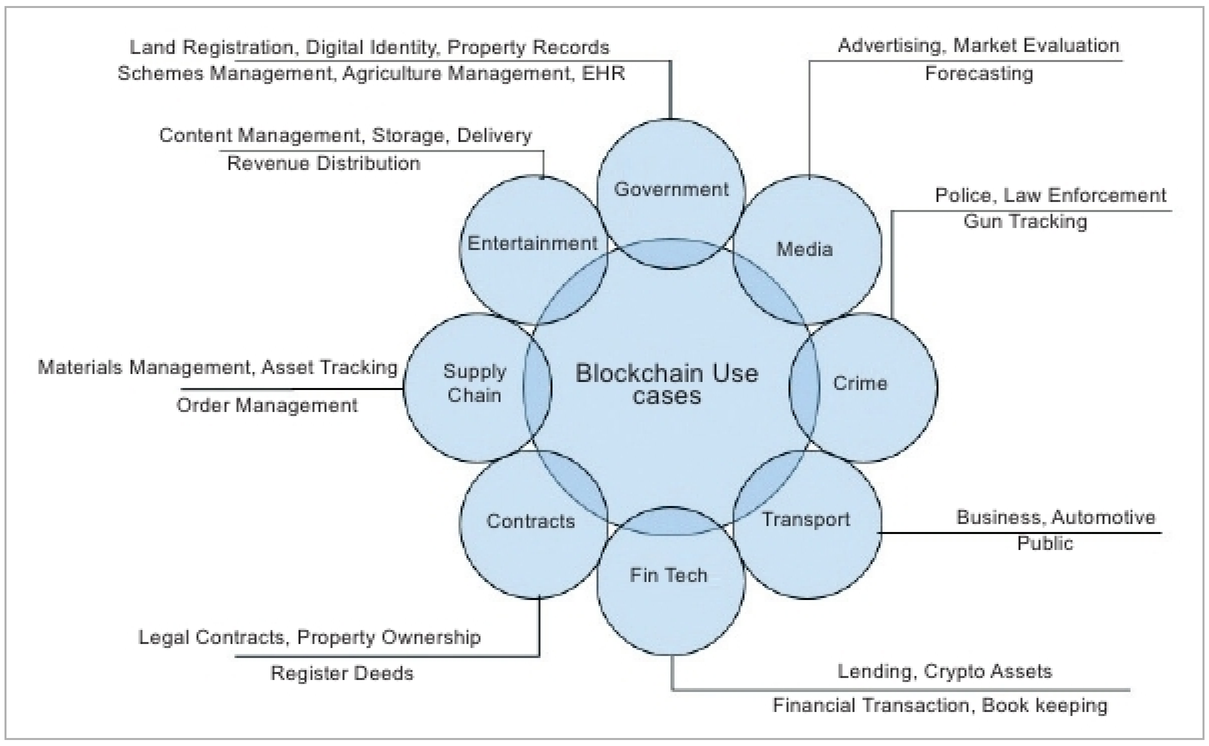


Figure 2: Blockchain use cases

In private blockchains, the owner is a single entity or an enterprise, which can override/delete commands on a blockchain if needed. It is not exactly decentralised, and is called a distributed ledger or database that uses cryptography to secure it.

A hybrid/federated blockchain should be able to connect a public blockchain open to every single person in the world, with a private blockchain running in a fully permissioned environment, i.e., limiting the access of available information.

Conducting business over a decentralised hybrid blockchain reduces transaction costs, eliminates data redundancy and speeds up transaction times.

Whether a blockchain is private or public, the user group that has the access to the information on that blockchain needs to be determined.

## Real world use cases of blockchain technology

Enterprises are going to take advantage of this new technology by optimising their network and automating it without the need for human interference. The real disruption is that the trust is established through collaboration and code, rather than a central authority. For example, we no longer need a bank to make a money transfer around the world. We no longer need an escrow account to buy a home, or a real estate agent to facilitate the transaction. With the penetration and adoption of blockchain technology, almost all the industries will get impacted. Figure 2 shows the blockchain use cases across various domains.

The main benefits of blockchain technology are:

- Greater transparency — transaction histories are becoming more transparent through the use of blockchain technology.
- Enhanced security — there are several ways the blockchain is more secure than other record-keeping systems.
- Elimination of error handling through real-time tracking of transactions with no double spending.
- Improved traceability.
- Trusted record-keeping and a shared trusted process.
- Increased efficiency and speed, and a reduction in settlement time to mere seconds by removing intermediaries.
- Reduced cost and complexity, and material cost reduction through the elimination of expensive proprietary infrastructure.
- Full automation of transactional processes, from payment through settlement.

However, the blockchain is not recommended for the following:

- High performance (millisecond) transactions